# SoK: Agent Communication Side Channels Executed Cover and Joint Privacy–State Evaluation

***Abstract*—Privacy cover in an Agent can become real tool execution. Its traffic, original-task outcome, business state and resource use must therefore be evaluated together. We systematize fifteen Agent communication-side-channel studies, comprising forty-four settings, against twenty-two traffic-analysis anchors. A complete main-text matrix organizes observations, targets and interventions. Four query/workflow privacy settings from two sources provide no setting with both an adapted attacker and matched original-task utility. We turn that evidence into a joint evaluation protocol and apply it to controller-generated and planner-generated cover. In 3,000 executions with two fixed-weight planners over MCP and TLS 1.3/HTTP/2, timing-based removal recognizes 112/120 and 108/120 immediate-cover tasks when call boundaries are supplied. Record-only attacks without those boundaries still recognize 91/120 and 97/120; removal depends on event reconstruction and model. Across the tested representations, shadow planners permit 73/120 and 69/120 recognition while approximately doubling model tokens. The evidence connects protection decisions to residual inference, execution authority, business invariants and the resources consumed by cover.**

## 1. Introduction

An Agent receives a support ticket, checks an order and submits a refund. An observer sees the contacted services, encrypted message sizes and gaps between requests. A defender adds decoy calls to obscure the real task. Those calls create two outputs: an observable communication trace and work performed by the application. A decoy may stop at a relay, execute on isolated state, or consume a business resource. Evaluating its traffic alone leaves the second output unmeasured.

The research question. How should Agent communication defenses be evaluated when cover is executable work? We systematize fifteen Agent studies, with forty-four settings, against twenty-two traffic-analysis anchors. The central contribution is a joint evaluation of residual inference, original-task completion, business state and resource cost under the same intervention. The evidence map identifies which existing results can populate that record; a controlled case applies it to controller-generated and planner-generated cover.

**A privacy result needs outcomes from the same protected execution.** WRA's published research decoys reduce query recovery from .770 to .565, while its report-utility scores 8.55 and 8.54 belong to a different intervention that changes destination visibility [2]. The communication-graph study refits an attacker on protected OHTTP captures and reports traffic cost, but its matched original-task utility is unresolved [3]. Across the four mapped query/workflow privacy settings from these two sources, none documents both attack adaptation and matched task utility. This is a concrete missing connection between measurements, not an estimate of how often the entire field omits it.

**The process generating cover is itself observable.** Honest tool calls follow model decisions; controller-inserted calls can skip that wait. In 3,000 executions with two fixed-weight planners over real MCP, TLS 1.3 and HTTP/2, given call boundaries, gap-based removal without cover-role labels restores recognition to 112/120 and 108/120. Historical-gap dispatch changes that shortcut, but adapted attacks remain effective. A complete shadow planner creates a different trace and approximately doubles model tokens. The case reports the strongest tested attacks alongside the frozen validation choices, and evaluates task, state and costs on the same runs.

Established WF and ML-security work already requires realistic observation units, protected-condition attacks and matched costs [15], [16], [52], [19]. Agent execution adds the need to specify where cover terminates, what privileges it has and whether its results return to the planner. Tool feedback also reaches later model requests, making service access part of protection scope. We use that relationship to organize residual views, rather than claim a new, isolated causal result from the inconclusive forwarding pilot.

Contributions.

- **An execution-aware field map**: all forty-four settings appear in a compact main-text matrix, with interfaces, observation rights, targets, signals, units, reported results and intervention evidence. The transfer matrix connects the Agent evidence to established communication analysis (Sections 2–4).
- **A joint privacy–business evaluation protocol**: cover termination and execution authority determine the state, task and resource outcomes that must accompany residual inference. A reusable checklist and source-matched examples make the protocol concrete (Sections 5, 6, 8).
- **A worked technical case**: actual controller and shadow dispatch, timing-aware removal, strongest tested inference and business outcomes expose which protection decisions each measurement supports. A record-only reanalysis removes supplied

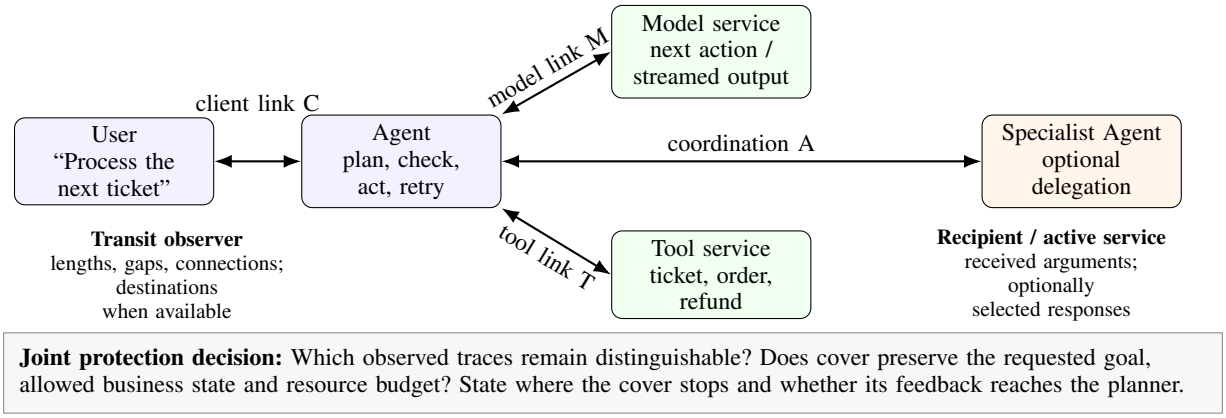



Figure 1. Running support-ticket example and observation rights. A tool result returns to the Agent and can enter the next model request. Each encrypted link exposes its own traffic shape; a recipient additionally sees the request addressed to it. Concurrent tickets may share a connection on either link. The arrows represent application communication, not a one-to-one mapping from tool events to packets.

call boundaries while retaining episode association (Section 7).

## 2. Scope and Evidence Construction

The evidence map connects Agent execution to an observed communication signal, a hidden target and the rights needed to observe it.

### 2.1. Inclusion and retrieval

An Agent uses an LLM to select actions, with external feedback able to affect later action or termination. Client, tool, model and inter-agent interfaces are eligible. Transit observers obtain traffic metadata; contacted recipients see their addressed requests; active auditors also choose probes or challenges. Negative results qualify: PrivacyTrace's metadata/order condition remains included despite scoring below its prior reference [35]. Whole semantic trajectories or browser instrumentation alone enter the adjacent layer, while deliberate covert messages and prompt-injection defenses without communication-side inference remain contextual related work [36], [26].

The fixed eligibility window covers first public submissions from 1 January 2024 through 9 November 2026. Twelve topic queries, a native arXiv query and inclusion rules preceded the retrieval pass; title/publication checks and one-step citation follow-ups completed it. The arXiv query returned 180 records. Public-web queries supplied 33 scholarly candidates across 101 displayed URLs; deduplication yielded 209 records. Two citation leads brought screening to 211: fourteen direct, forty adjacent, 155 outside scope and two unresolved identities. The eligible CHeaT (Cloak, Honey, Trap) seed adds the fifteenth direct source (Figure 2). The SSH honeypot entered through citation tracing, illustrating the value of combining retrieval routes. Exact queries and dispositions are in the supplement.

Twenty-two separately selected mechanism anchors comprise sixteen published communication studies and six

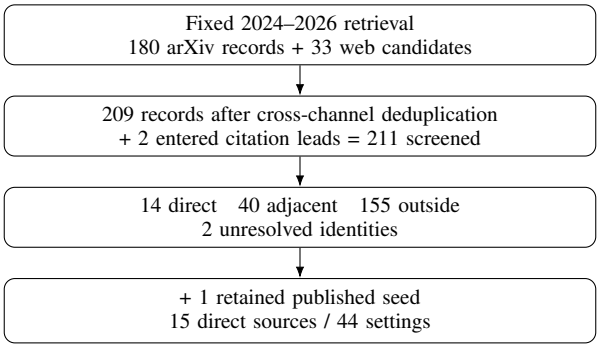



Figure 2. Executed coverage audit. Counts are deduplicated work/lead records; two identities remain unresolved. Mechanism anchors use a separate inclusion rule.

serving studies. They supply the established relations in the transfer matrix; the fifteen direct studies supply the Agent application evidence. These layers retain separate inclusion records and denominators. The anchor set is selected for mechanism correspondence, rather than a census of classical traffic analysis.

### 2.2. Extraction and source dependencies

Forty-four setting records bind source version and locator to access, features, learner, target, population, split, intervention, result and cost. Labels, execution outcomes, failure treatment and policy context are recorded alongside them. WRA and CHeaT are checked against official USENIX texts. One HTTP study has a verified ACM workshop identity, while its detailed measurements come from the extended arXiv version [49], [50]. Twelve other works lack independently verified formal publication identities at the cutoff.

Text reading, displayed-number reconstruction, selected-code inspection and data recomputation have separate entries in the extraction sheet. For example, WRA's decoy and visibility conditions are checked in the published text; OHTTP

refitting is checked in selected code. Each check supports its matched premise. Section 9.1 reports what remains when sources are removed. The author assigned and decided every code against the cited passage. AI tools located candidate passages and proposed draft codes, which the author accepted, changed or rejected. Two further AI passes audited the codebook; they are not inter-rater reliability. An independent security PhD researcher, blind to the manuscript and author codes, re-coded the decision-critical subset from the sources. Access class (44/44), intervention purpose (9/9), and adaptation and utility in the four privacy settings (4/4 each) agree fully ($\kappa = 1$). The protection flag agrees on 38/44 ($\kappa = .61$) and transfer relations on 5/6 ($\kappa = .77$); after source adjudication of the seven disagreements, no setting has both an adapted attacker and matched utility (Supplement S6.1). The source-located table in Supplement S7 lists each privacy decision. The low-agreement execution-outcome field enters only the descriptive coverage summary in Supplement S6.

For the descriptive summary, every setting receives an access class and intervention purpose. Protected-condition adaptation and original-task utility are counted within the four query/workflow privacy settings. Three identity-obfuscation and two adversarial-Agent blocking settings have different objectives; one composition setting remeasures CG-transform's protections and is counted with it; thirty-four settings contain no protection comparison. A located item identifies evidence in the retained record. An unlocated item leaves that question open. The supplementary reporting ledger lists the assignments and source locators.

### 2.3. Relation to security surveys and Agent defenses

Broader surveys organize LLM and Agent threats by lifecycle, attack surface and defensive role [53], [54], [55]. This SoK asks three communication-specific questions: which service representation reaches which observer, whether an intervention covers the later feedback path, and what business work its cover can trigger. The maps connect each answer to a measured setting and its evidence strength. The supplementary deployment section compares these surveys and WF systematizations.

Information-flow and privilege controls address closely related execution boundaries. CaMeL tracks dependencies and explicitly discusses tool-call and timing side channels [57]; FIDES enforces confidentiality/integrity policies and selectively exposes tool information [58]; Progent constrains tool calls and arguments [59]. SPILLage distinguishes content and behavioral oversharing by browser Agents [27]. Our maps add the communication-observer question: which metadata remain visible under a particular deployment and intervention? They do not replace these policy mechanisms. Likewise, $\tau$-bench and AgentDojo already evaluate environment outcomes [60], [61]. The cover example applies state-aware evaluation to matched traffic measurements; state checking itself is established practice.

## 3. Threat Model and Classification

**A result is meaningful only for the target and observations actually supplied to its attacker.** The ticket example has client–Agent, Agent–model, Agent–tool and coordination links (Figure 1). A transit observer sees connection destinations, directions, lengths and times. A contacted service reads the request addressed to it; an active auditor can also choose challenges. Table 1 maps these rights to deployment configurations. The same traffic may encode requested intent, a logged action or successful execution. Authorization additionally requires the principal and policy context: identifying a refund does not decide whether it was approved.

### 3.1. What is observed and what is predicted

A captured packet, TLS record, HTTP stream, tool call and complete task are different units. A gateway can regroup streamed output; HTTP/2 and QUIC can multiplex streams; MCP may use local stdio or remote HTTP [23], [24], [39], [40], [25]. A boundary supplied by the evaluator is prior information, not a classifier output. Six fields specify each inference setting: interface and representation, observer access, supplied prior information, prediction target, evaluation population and base rate, and protected-training access. Host traces can supply experimental truth without becoming transit-visible features [30], [31].

### 3.2. Why execution matters to cover

Transport cover can end before a business tool. Executed cover crosses an application boundary: it may consume quota, access other objects, update state or enter the planner's context. We classify a policy by its termination point, state isolation, feedback visibility and dispatcher. These choices determine both its observable signal and the original task's execution budget. Information forwarded to another service creates another permitted observation; it does not imply that every service is visible to the same attacker.

## 4. Evidence Map and Classical Conditions

**Most mapped settings evaluate recognition; only four settings from two sources support the selected query/workflow privacy comparisons.** Tables 2–3 display all forty-four settings. They preserve the source-defined target and unit rather than rank incompatible accuracy, reconstruction and prevention scores. Twenty-two settings use transit observations, ten use recipient access, eleven active access, and one inferred preferences. Three settings compare identity obfuscation and two compare adversarial-Agent blocking; these are separate from user/workflow privacy.

### 4.1. What transfers from communication analysis

Table 4 separates established evaluation conditions from additional Agent execution conditions. WF already ad-

TABLE 1. DEPLOYMENT AND ACCESS MAP. THE CONFIGURATIONS ARE ANALYST-DEFINED; CITED SETTINGS TEST THE LISTED VIEW, NOT EVERY COMPONENT OF THE DEPLOYMENT. A GATEWAY'S PLAINTEXT ACCESS REQUIRES TLS TERMINATION.

| Deployment | Observer and permitted view | Existing evidence for that view | Protection consequence |
|---|---|---|---|
| Enterprise egress | A transit monitor sees routed model/tool flows, sizes, times and any exposed destination association. | WRA: supplied research-domain traces; ANT and cloud benchmark: model/combined traffic [2], [9], [4]. | Specify which services cross the monitored egress. A union of flows is available only when the monitor can observe them. |
| Shared gateway or relay | Transit sees multiplexed gateway output. A terminating gateway can additionally read addressed requests. | Live OHTTP: passive-link inference under stated relay trust [3]; PrivacyTrace: semantic recipient/gateway access [35]. | Separate relay association protection from content trust. Connection sharing changes the observation unit. |
| Local model, remote tools | An external observer sees remote tool traffic; the local model call has no corresponding external network leg. | WRA's destination-recovery mechanism is relevant to tool egress [2]; whole-deployment protection is untested in the mapped comparison. | Evaluate tool and downstream calls. Do not import a remote-model attack score into an absent network leg. |
| Cloud model, remote tools | An authorized vantage may see either or both encrypted legs. A contacted tool sees its own arguments. | ANT/cloud benchmark test model traffic [9], [4]; PrivacyTrace tests recipient-visible representations [35]. | Check permitted model and tool views after the intervention; recipients and passive monitors need different protections. |

dresses open-world populations, base rates, unknown boundaries and adaptive attacks [15], [45], [43], [44], [16]. Web and IoT work connects action-dependent messages to observable events [28], [29], [33], [22]. Agent execution couples those events to model waits, tool state and later feedback. A condition is retained when it applies directly, extended when an execution condition must be added, and untested when the mapped Agent evidence does not evaluate the corresponding claim. These are relationships, not maturity levels.

### 4.2. Changes of execution are distinct tests

Communication-graph composition holdout keeps nine known labels and changes their capability combinations: reported accuracy moves from about .67 to .18 [3]. A cloud benchmark's graph/star comparison instead changes coordination, with about .83 to .52 under its matched feature protocol [4]. Trace holds out Agent scaffolds for model-family attribution, reporting macro-F1 .815 versus .981 in a mixed-session test [48]. These are version-specific observations on different targets. A pooled transfer score would erase the condition each tests. Failures also select the population: first-pass attribution, repaired trajectories and successful-run filtering answer different questions.

## 5. Executed Cover Changes the Evaluation Contract

**A deployable cover policy must meet both its inference target and the business constraints of the original task.** The map supplies two practical distinctions: evidence from different interventions cannot be joined into a single operating point, and original-task completion covers only part of application utility.

### 5.1. Protection and utility must name the same intervention

WRA's five research decoys lower functional query recovery from .770 to .565. Its 8.55/8.54 report-utility comparison instead changes destination visibility from 100% to 10%, with recovery .770/.610 [2]. Those utility scores do not price the executed-decoy result. This distinction blocks a concrete interpretation: the pair cannot establish that five decoys preserve report quality at that measured level.

The communication-graph live OHTTP setting reports six-class accuracy .30, approximately thirteen-fold bytes and eleven-fold latency, with protected-capture RF refitting [3]. Its nine-class .67 workflow result is a different population. Traffic cost is recorded, while original-task utility under that intervention remains unresolved. Table 9 gives each eligible setting and its evidence. Among four settings, adaptation is documented in two and matched task utility in one; no setting documents both. The count identifies a gap in these inspected configurations.

### 5.2. Business state is separate from goal completion

A goal predicate can hold while another order, shared quota or ledger is wrong. A joint record therefore separates requested-goal success, violations of declared business invariants, and resource use. In production, invariants should allow legitimate concurrent updates; exact state equality is appropriate for a fixture whose intended transition is fully specified. State-aware evaluation already exists in $\tau$-bench and AgentDojo [60], [61]. Our contribution connects it to the same traffic-protection intervention and adapted observations.

An execution contract names the allowed objects and transitions, quota/token budgets and feedback that cover may

TABLE 2. COMPLETE SETTING MATRIX, PART 1 OF 2. EACH ROW IS ONE OF THE FORTY-FOUR RETAINED SETTINGS. SCORES KEEP THE SOURCE'S METRIC AND TARGET; THEY ARE NOT CROSS-PAPER RANKINGS. ACCESS STATES THE OBSERVER'S RIGHTS. LAST COLUMN GIVES INTERVENTION PURPOSE AND, FOR THE FOUR PRIVACY COMPARISONS, ADAPTED ATTACK / SAME-INTERVENTION ORIGINAL-TASK UTILITY (YES OR NR: NOT LOCATED). A DASH IS OUTSIDE THAT DENOMINATOR. SHARED* IS COUNTED WITH CG-TRANSFORM. FULL FEATURES, SPLIT RULES AND LOCATORS REMAIN IN THE EXTRACTION SHEET.

| Work / setting | Link; access | Target | Signal / learner | Evaluation unit | Source result | Purpose; A/U |
|---|---|---|---|---|---|---|
| AP-identity [1] | client; transit | hosted-agent identity | Packet size/direction; CNN | interaction | F1 .8660 closed; .8477 open | none; – |
| AP-behavior [1] | client; transit | interaction mode | Packet size/direction; CNN | interaction | F1 .924/.889 conflict | none; – |
| AP-persona [1] | preferences; derived | occupation | Inferred preferences | participant preferences | Top-3 occupation 69.1% | none; – |
| WRA-prompt [2] | tool/Web; transit | prompt meaning | Domains/time; LLM | research task | Query recovery .770 / .725 | none; – |
| WRA-persona [2] | tool/Web; transit | user traits | Domains/time; LLM | multiple sessions | Separate trait-category scores | none; – |
| WRA-decoy [2] | tool/Web; transit | prompt meaning | Domains/time; LLM | research task | .770 to .565; five decoys | privacy; NR/NR |
| WRA-visibility [2] | tool/Web; transit | prompt meaning | Domains/time; LLM | research task | .770/.610; quality 8.55/8.54 | privacy; NR/yes |
| CG-workflow [3] | coord.; transit | workflow class | Graph/size/time; RF | workflow | Nine-class accuracy .67 | none; – |
| CG-composition [3] | coord.; transit | workflow class | Graph/size/time; RF | workflow | Known-label holdout .18 | shared*; – |
| CG-MCP [3] | tool; transit | sequence class | Graph/size/time; RF | fixed sequence | Fixed-sequence accuracy .99 | none; – |
| CG-transform [3] | coord.; transit | workflow class | Graph/size/time; RF | workflow | Capture transformation; separate from live | privacy; yes/NR |
| CG-live [3] | coord.; transit | workflow class | Graph/size/time; RF | workflow | Six-class .30; 13x bytes, 11x delay | privacy; yes/NR |
| CG-active [3] | coord.; active | integrity impact | Graph/size/time; RF | pending workflow | Live integrity endpoint | none; – |
| CB-task [4] | model; transit | task | Traffic statistics; RF | MARBLE execution | Ten-class BA .853/.860 | none; – |
| CB-topology [4] | model; transit | task | Traffic statistics; RF | execution | Topology change .83 to .52 | none; – |
| CB-size-excluded [4] | model; transit | task class | Traffic statistics; RF | execution | BA .540/.587 | none; – |
| CB-partial [4] | model; transit | task | Traffic statistics; RF | execution | One-agent BA .665/.780 | none; – |
| PF-session [5] | tool/Web; transit | synthetic persona | Signed intervals; CNN | fixed packet window | Per-site persona F1 .78–.91 | none; – |
| PF-pooled [5] | tool/Web; transit | synthetic persona | Signed intervals; CNN | packet window | Pooled summaries conflict | none; – |
| PF-open [5] | tool/Web; transit | persona membership | Signed intervals; CNN | window | Open-world misattribution | none; – |
| MARK-protocol [6] | tool/Web; recipient | agent stack identity | Protocol/headers | trial | TLS/H2 F1 .725; HTTP .799 | none; – |
| MARK-combined [6] | tool/Web; recipient | agent identity | Protocol/headers | trial | Combined F1 .971 | none; – |

expose to the real planner. Isolation prevents direct cover writes to production state but still consumes compute and can change scheduling. A response being hidden from the planner does not undo the underlying operation. Section 7.1 evaluates these outputs separately.

### 5.3. Protection scope includes other permitted views

WRA observes research destinations, whereas ANT and the cloud benchmark measure tasks on model connections [2], [9], [4]. A recipient additionally sees its addressed arguments: PrivacyTrace reports attribute-inference ASR .2771 with arguments, versus .1453 for metadata/order and a .1797 schema reference [35]. This is an access distinction, not a ranking across attackers. Forwarded tool results enter later model requests, alongside initial task fields and model decisions. A tool-only defense must therefore state which other views its attacker retains. The separate forwarding pilot has non-rejecting primary tests and does not isolate the causal contribution of feedback content (Supplement S10).

## 6. Defense Mechanisms and Execution Costs

**The layer that controls a signal determines which protection and business costs it can change.** Table 5 classifies naming, representation, scheduling, transport cover, execution cover and shared-state isolation. The map extends traffic-budget comparisons with tool privileges, business state and model work.

Name hiding changes destination association; message padding changes lengths; pacing changes timing and termination. STP's tunnel endpoints discard cover before devices process it [34]. Walkie-Talkie, FRONT, Adaptive Padding and WTF-PAD supply different scheduling and overhead choices [17], [18], [63], [64]. Their evaluation lessons carry over, while executed research or tool calls add application work. Matching interarrival marginals alone is not a

TABLE 3. COMPLETE SETTING MATRIX, PART 2 OF 2. EACH ROW IS ONE OF THE FORTY-FOUR RETAINED SETTINGS. SCORES KEEP THE SOURCE'S METRIC AND TARGET; THEY ARE NOT CROSS-PAPER RANKINGS. ACCESS STATES THE OBSERVER'S RIGHTS. LAST COLUMN GIVES INTERVENTION PURPOSE AND, FOR THE FOUR PRIVACY COMPARISONS, ADAPTED ATTACK / SAME-INTERVENTION ORIGINAL-TASK UTILITY (YES OR NR: NOT LOCATED). A DASH IS OUTSIDE THAT DENOMINATOR. SHARED* IS COUNTED WITH CG-TRANSFORM. FULL FEATURES, SPLIT RULES AND LOCATORS REMAIN IN THE EXTRACTION SHEET.

| Work / setting | Link; access | Target | Signal / learner | Evaluation unit | Source result | Purpose; A/U |
|---|---|---|---|---|---|---|
| FAY-network [7] | tool/Web; recipient | implementation | Protocol/site features | connection/request | Layer-specific distinguishability | identity; – |
| FAY-combined [7] | tool/Web; recipient | agent identity | Protocol/site features | visit | Combined fingerprint view | identity; – |
| FAY-blocking [7] | tool/Web; active | access outcome | Protocol/site features | visit | Access success; not privacy score | blocking; – |
| AT-choice [8] | tool/Web; active | cross-origin fact | Challenge choice | two-origin session | 61.1% loaded choice; cold-adjusted LS | none; – |
| ANT-scenario [9] | egress; transit | scenario | Direction/size/time | episode | Five-task accuracy .431–.727 | none; – |
| ANT-model-scope [9] | model/rest; transit | scenario | Direction/size/time | episode | Eight approximate model-view scores | none; – |
| ANT-risk [9] | egress; transit | benchmark risk label | Direction/size/time | episode | 209 episodes; risk recall .15–.95 | none; – |
| ANT-stage [9] | egress; transit | primitive group | Direction/size/time | primitive segment | 47-group accuracy .355–.462 | none; – |
| CHT-lure [10] | asset; active | LLM-based visitor presence | Lure/output/challenge | probe interaction | Presence mechanism; no natural F1 | none; – |
| CHT-password [10] | output; active | LLM presence | Lure/output/challenge | challenge response | Password-output control | none; – |
| CHT-blocking [10] | CTF; active | defender prevention success | Lure/output/challenge | CTF attempt | CTF prevention endpoint | blocking; – |
| PT-metadata [35] | tool; recipient | profile attributes | Tool metadata/arguments | single-domain trajectory | ASR .1453; prior .1797 | none; – |
| PT-content [35] | tool; recipient | profile attributes | Tool metadata/arguments | single-domain trajectory | Arguments ASR .2771 | none; – |
| ATR-main [47] | tool; active | model identity | Arguments/repairs | multi-probe attribution trial | 50-probe accuracy .960 | none; – |
| ATR-budget [47] | tool; active | model identity | Arguments/repairs | multi-probe attribution trial | 1–50 probes: .652–.960 | none; – |
| ATR-obfuscation [47] | tool; active | model identity | Arguments/repairs | multi-probe attribution trial | Attribution above .94 | identity; – |
| TR-within [48] | terminal; recipient | model family | Terminal n-grams; SVM | terminal session | Mixed-session F1 .981 | none; – |
| TR-scaffold [48] | terminal; recipient | model family | Terminal n-grams; SVM | terminal session | Scaffold holdout F1 .815 | none; – |
| TR-proprietary [48] | terminal; recipient | model family | Terminal n-grams; SVM | terminal session | Accuracy .78; narrower models | none; – |
| HTTP-signature [49] | docs; recipient | implementation | HTTP trace | documentation-retrieval trial | 9 agents/6 services; descriptive | none; – |
| HP-calibration [51] | SSH; active | Agent candidate | Timing/prompt responses | controlled probe interaction | Probe timing about 1.7s | none; – |
| HP-field [51] | SSH; active | Agent candidate | Timing/prompt responses | field interaction/session | 8 candidates; identity unverified | none; – |

new security principle: receive-dependent padding already models request–response dependencies. The Agent-specific object tested here is the controller/model decision process that generates tool calls.

Model services introduce additional timing and sharing signals. Token streams, cache behavior and token count can remain observable under particular interfaces [20], [12], [11], [13], [21], [14]. A tool-message padding rule does not determine a later model request's shape. In the separate concurrent pilot, one constant-rate policy increases model bytes 19.5–22.5 times without a consistent inference improvement; this is a measured implementation, not a verdict on all pacing designs. Defense comparisons should report the achieved downstream trace, task/state outcomes and resources together.

## 7. A Joint Evaluation of Executed Cover

**Cover generation changes residual inference and resource use, while business correctness requires a separate endpoint.** The five-arm live study compares dispatch policies on actual TLS records. A record-only extension withdraws call boundaries; a separate earlier state control tests whether completion alone can miss collateral effects. These populations and their statistical roles remain separate.

TABLE 4. TRANSFER OF ESTABLISHED CONDITIONS TO AGENT EXECUTION. RELATIONSHIP LABELS CONCERN THE STATED MAPPING; EVIDENCE STRENGTH AND MISSING TESTS APPEAR SEPARATELY. A MISSING AGENT TEST DOES NOT REFUTE THE CLASSICAL RESULT.

| Established relation and condition | Agent mapping / consequence | Relation | Agent evidence and remaining test |
|---|---|---|---|
| Unknown boundaries and background affect event recognition [45], [43], [44], [33]. | Tool bursts, model waits and task intervals are different units. Evaluate proposed events with false alarms as well as classification. | Extended | ANT supplies stage boundaries; the local concurrent study tests particular detectors. Greater Agent localization difficulty is untested. |
| Defense evaluation must allow adaptation to the protected distribution [32], [16], [19]. | Record protected training access and still-visible links; an unchanged learner family can refit. | Retained | Transformed and live OHTTP settings report adaptation in one preprint [3]; WRA's matched protected examples remain unresolved. |
| Coarse timing/volume can remain useful under specified padding [32]; serving streams expose length/gap signals [20], [12]. | Name the shaping policy, receiving hop and Agent workload before carrying over a measured effect. | Untested generalization | Cloud and ANT measure model-view recognition [4], [9]. A general protected-Agent SSE claim is not tested by these observations. |
| An observation result is tied to the visible paths and representations [28], [34]. | Feedback forwards tool information into later model requests across separately controlled services; evaluate every permitted residual view. | Extended | Published WRA tool-side evidence and preprint model/recipient results establish different exposed interfaces. Causal feedback attribution needs its own test. |
| STP tunnel endpoints remove cover before it reaches devices or servers [34, Sec. 7.1]. | When cover executes as research, add original-task resource, state and feedback constraints to the traffic budget. | Extended | Published WRA reports executed research decoys and utility under a different intervention [2]. Our controlled LLM case separates task completion from state integrity (Section 7.1); it does not evaluate WRA writes. |
| Host traces can reconstruct distributed request paths [30], [31]. | Instrumentation can provide ground truth; a transit attacker must infer from its own permitted observations. | Access mismatch | ANT stage labels and local task intervals are evaluator-provided. Host tracing alone cannot establish passive encrypted-stream localization. |

## 7.1. Actual dispatch and timing-aware attacks

**Immediate controller cover is removable because it can skip the model wait preceding honest calls.** Two fixed-weight planners, Qwen3:8B and Llama3.1:8B, each execute 300 fixtures in five conditions, yielding 3,000 main episodes and 600 additional shadow chains. The six business actions use seven MCP services through LangGraph over TLS 1.3/HTTP/2. Original partitions contain 120 training, 60 validation and 120 paired test fixtures per cell. The task label is requested intent; failures remain included. Cover is isolated from real business state and hidden from the real planner.

No cover establishes the baseline. Immediate cover dispatches after an honest response. Historical sequential and concurrent policies sample training-only honest gaps and service durations, wait in real time and hold responses for the sampled minimum duration. The concurrent dispatcher overlaps real model inference. Shadow runs a complete second planner on an independent six-class ticket; either chain can start first. Calls to the same service serialize. The original observer receives destinations, ciphertext bytes, TLS record completion times and supplied episode/call windows. Section 7.2 removes the last of these call-level privileges.

Removal and adaptation. An unlabelled training-gap mixture selects the short-gap component for removal; a no-cover-trained task classifier decodes the remaining trace. In the immediate arm this restores 112/120 and 108/120 correct predictions, versus selected non-timing references 75 and 80. Both meet the prespecified replication criterion and pass Holm correction. Independent TCP reassembly matches all 773,450 recorded TLS record types, lengths and counts. Recomputing the GMM attack with packet-capture clocks reproduces the scores; software completion timestamps retain measured error (Supplement S2).

Historical sampling changes the short-gap attack, but adapted context removal or joint decoding still recognizes 93/120 and 104/120 sequential executions. The Qwen comparison with its 76/120 non-timing reference has Holm $p = .0559$; Llama's comparison with 82/120 has $p = .00078$. Concurrent adapted counts are 99/120 and 106/120, against 66 and 68. Their 27.5- and 31.7-point *attacker advantages under the same policy* both pass correction. The suite maxima in Table 6 additionally expose Qwen's sequential joint decoder at 99/120, rather than replace the frozen selection after testing.

The policy samples minimum waits, not achieved timing distributions. Qwen's mean 40.5 ms sampled hold becomes 100.4 ms sequential and 171.2 ms concurrent at the record observer. A failed simple mixture is therefore not evidence

TABLE 5. DEFENSE DESIGN SPACE: ENFORCEMENT AUTHORITY, COSTS TO MEASURE AND REMAINING EXPOSURE. SOURCE-SPECIFIC MEASUREMENTS APPEAR IN THE TEXT AND SUPPLEMENT S7.

| Layer / mechanism | What must be trusted or controlled | Costs to measure together | Residual exposure and practical decision |
|---|---|---|---|
| Naming / association: encrypted name resolution, relays, OHTTP | Name resolver or relay placement; the parties able to link the user to the destination | Added hops, bytes, latency and availability | Shape can survive name hiding. Choose this layer when destination association is the private target; then test remaining observers. |
| Representation: message, TLS-record or packet padding | Component that controls the final visible size; padding survives later encoding | Added bytes, record/packet count, fragmentation | Timing, count and other links remain. State the padding layer and test downstream captures. |
| Scheduling: batching, pacing, fixed-rate cover | Sender/relay controls output schedule and cover lifetime | First-response delay, completion delay, bandwidth during idle periods | Admission, termination and overflow can reveal activity. Budget these events as part of the visible trace. |
| Transport cover: dummy exchanges discarded at a tunnel endpoint | Trusted cover terminator; no dispatch into business tools | Cover bandwidth, queueing, resource use | Other endpoints still see their inputs. Prefer this boundary when the aim is traffic shaping without extra business work. |
| Execution cover: extra research or tool calls | Tool privileges, isolated state and whether decoy feedback reaches the planner | Model tokens, tool quota, task success, state changes and latency | A decoy can affect the real execution. Measure privacy and original-task utility on the same protected runs. |
| Shared-state isolation / minimization | Tenant separation, cache keys and information passed between services | Lost cache reuse, compute, latency, task accuracy | Isolation removes a sharing channel, not every traffic signal. Minimize tool arguments and feedback according to each recipient's role. |

TABLE 6. JOINT EVALUATION OF THE FIVE LIVE POLICIES, 120 TEST FIXTURES PER ROW; UNIFORM SIX-CLASS REFERENCE 20/120. SEL. IS THE VALIDATION-SELECTED NON-TIMING/ADAPTIVE COUNT; MAX IS THE LARGEST TEST COUNT IN THE FROZEN SUITE, REPORTED DESCRIPTIVELY, NOT SUBSTITUTED INTO THE PRIMARY TESTS. GMM IS ROLE-LABEL-FREE REMOVAL. BYTES GIVES WHOLE-CONNECTION/CALL-WINDOW FACTORS; TOKENS INCLUDES BOTH PLANNERS. GOAL AND STATE ARE SEPARATE COUNTS. IN SHADOW, BOTH SELECTORS CHOOSE THE IDENTICAL NON-TIMING ATTACK.

| Planner | Policy | Sel. S/A | Max | GMM | Bytes× | Tokens× | Goal/State |
|---|---|---|---|---|---|---|---|
| Qwen3:8B | No cover | 117/117 | 117 | – | 1.00/1.00 | 1.00 | 113/114 |
| Qwen3:8B | Immediate | 75/112 | 112 | 112 | 1.15/1.33 | 1.00 | 111/112 |
| Qwen3:8B | History, sequential | 76/93 | 99 | 53 | 1.15/1.32 | 1.00 | 112/113 |
| Qwen3:8B | History, concurrent | 66/99 | 99 | 38 | 1.15/1.33 | 1.00 | 111/112 |
| Qwen3:8B | Planner shadow | 54/54 | 70 | 34 | 1.31/1.68 | 2.01 | 112/113 |
| Llama3.1:8B | No cover | 110/110 | 112 | – | 1.00/1.00 | 1.00 | 82/70 |
| Llama3.1:8B | Immediate | 80/108 | 108 | 108 | 1.13/1.27 | 1.01 | 83/67 |
| Llama3.1:8B | History, sequential | 82/104 | 104 | 14 | 1.13/1.26 | 1.01 | 84/67 |
| Llama3.1:8B | History, concurrent | 68/106 | 106 | 66 | 1.13/1.27 | 1.02 | 82/67 |
| Llama3.1:8B | Planner shadow | 64/64 | 64 | 43 | 1.35/1.71 | 1.99 | 82/67 |

of successful concealment. In the saved packet-clock diagnostic, Llama's sequential mixture deletes 477/634 honest calls while detecting only 64/182 cover calls, explaining its 14/120 task count. Adapted joint decoding on the same condition reaches 104/120.

Planner-generated cover and cost. In the call-window suite, shadow's strongest attack is the size/destination RF: 70/120 Qwen and 64/120 Llama, or 58.3% and 53.3%, versus the 16.7% uniform reference. A chain-separation attack scores 49 and 34 and exactly recovers the real chain in 29 and 32 fixtures. The validation selectors choose the same non-timing attack for both roles; their equality is not an independent role-indistinguishability test. Shadow uses 2.01/1.99 times model tokens, 1.68/1.71 times call-window bytes and mean real-chain times 4.80/3.94 s versus baseline 2.91/3.33 s. It is a measured operating point, not an equal-budget demonstration that planning itself is superior. Supplement S2 retains every attack, primary test and cost.

## 7.2. Recognition without supplied call boundaries

**Withdrawing call endpoints tests a different attacker representation, while whole-episode association remains available.** We reanalyze all 3,000 recorded episodes using complete encrypted connections, including initialization and closing records. The attacker receives only relative record times, directions, ciphertext lengths and destination identities. It receives no call starts/ends, operation names,

TABLE 7. NO SUPPLIED CALL BOUNDARIES. ALL ENCRYPTED RECORDS IN ASSOCIATED EPISODE CONNECTIONS ENTER THE ANALYSIS. CELLS GIVE SELECTED IDLE-GRID BASELINE / REMOVAL / LARGEST TEST COUNT INCLUDING THE LATER EXPLORATORY PROTOCOL-AWARE SUITE, OUT OF 120; UNIFORM REFERENCE20. WHOLE-TASK ASSOCIATION REMAINS SUPPLIED.

| Policy | Qwen | Llama |
|---|---|---|
| No cover | 115/109/117 | 108/101/112 |
| Immediate | 80/69/91 | 90/86/97 |
| History, sequential | 73/74/75 | 88/56/93 |
| History, concurrent | 56/61/62 | 78/62/88 |
| Planner shadow | 69/65/73 | 69/56/69 |

role labels or HTTP stream assignment. A per-connection idle-gap rule proposes events; five gaps from 5 to 250 ms compete on validation task accuracy. Raw-record classifiers bypass this eventization entirely.

For each condition, LR and RF use destination summaries and early/late event sequences, with and without timing. A training-only GMM removes short-gap events before a classifier trained on no-cover traces. Validation separately chooses the baseline and removal attacks. This is a retrospective, fixed-protocol analysis of reused fixtures; only the immediate-arm paired comparison is a primary test, with Holm over the two models. Table 7 also gives each finite suite's largest test count without substituting that maximum into the primary test.

On immediate cover, selected removal recognizes 69/120 Qwen and 86/120 Llama tasks, against baselines 80 and 90. The paired differences are -9.2 and -3.3 points (Holm $p = 0.02545$ and $0.5413$). Removal underperforms the baseline in Qwen after correction; Llama does not reject. The original removal advantage is not reproduced by this boundary-free idle-gap pipeline. After observing the idle-grid results, an exploratory protocol-aware eventizer uses the public request-padding quantum and MCP setup sequence: large upward records mark proposed requests, and large downward records delimit responses before the next request. It does not read call truth. Within this representation, selected removal changes Qwen recognition from 68 to 91, but Llama from 90 to 87; these exploratory counts do not replace the original tests. Immediate cover permits maxima 91/120 and 97/120 across the record-only suites. Whole-episode association remains given, and differing representations prevent interpreting the change from the earlier call-window count as a pure causal boundary effect. Supplement S3 gives the full protocol and independent event-matching diagnostic.

The independent event audit gives protocol-event F1 0.295 and 0.380 at IoU .5 on immediate cover. Task recognition from record summaries therefore does not imply accurate recovery of calls.

### 7.3. Business constraints and the protection decision

Every live cover arm executes in isolated state and hides cover feedback from the real planner. Table 6 reports goal/state counts on the same fixtures as inference and costs. Shadow outcomes are 112/113 for Qwen and 82/67 for Llama, versus no-cover 113/114 and 82/70. These are different endpoint counts: a correctly preserved state need not contain a successful intended action. Low Llama baseline performance and changed planning paths limit business-equivalence conclusions.

The earlier six-policy study supplies a deliberately adverse state control (Supplement S10). Read/proxy/hidden-isolated cover retains Qwen state correctness at 115–116/120. Live writes can alter other objects and reduce it to 7/120; 95 of 102 completed Qwen tasks have collateral errors. Llama's corresponding counts are 3/120 intact and 73 of 76 completed tasks with collateral errors. The harm direction is built into that stress condition; its role is to test whether the evaluation records forbidden effects, not estimate production harms or reproduce WRA decoys.

The combined decision is concrete. Immediate cover leaves high inference after removal. Historical dispatch changes that attack but retains substantial adapted inference. Shadow execution trades additional model work for a different observed trace, with residual task recognition. An adverse write can fail the state contract even when the requested goal passes. Choosing a policy requires stating acceptable inference, task, state and resource tolerances rather than comparing a single accuracy decrease.

## 8. A Reusable Joint Evaluation Protocol

**Privacy, task success, business state and cost must refer to the same intervention and evaluation population.** Table 8 is a compact reporting form; the procedure below explains how to use it. It combines established security-evaluation guidance [52], [15], [16], [19] with explicit cover execution constraints. It is a reporting proposal, not a measured analyst-performance improvement.

1) **Specify observations and execution authority.** Name the hidden target, visible links, recipient rights and supplied boundaries. State where cover ends, which state it can access and whether its responses reach the planner.
2) **Specify business invariants and budgets.** Define the original goal, allowed collateral transitions and quota/token/latency tolerances. Separate a goal that was attempted, one that completed and a whole execution satisfying its state contract.
3) **Measure matched executions and adapted attacks.** Pair baseline and policy on the same workload. Include cover-detection/removal and schedule-aware attacks, plus the largest tested residual inference. Report original-task outcomes and all resource costs from those protected runs.

TABLE 8. JOINT EVALUATION CHECKLIST. COMPLETE ALL APPLICABLE ENTRIES FOR ONE INTERVENTION AND POPULATION; ATTACH EVIDENCE OR STATE THE MISSING ITEM. GOAL COMPLETION AND WHOLE-STATE CORRECTNESS HAVE SEPARATE FIELDS.

| | Entry | | Entry |
|---|---|---|---|
| ☐ | Hidden target, population, prior/base rate | ☐ | Allowed observations and supplied boundaries |
| ☐ | Cover termination, privileges and isolation | ☐ | Planner-visible cover feedback and dispatcher |
| ☐ | Adapted attacks and largest tested residual | ☐ | Original-task attempts, completion and failures |
| ☐ | Business invariants and violations | ☐ | Matched bytes, latency, tokens and quota |

TABLE 9. EVIDENCE LOCATED IN THE RETAINED RECORDS. COUNTS USE ELIGIBLE SETTINGS; THE SECOND COLUMN ALSO GIVES DISTINCT SOURCES. "NOT LOCATED" LEAVES EVIDENCE UNRESOLVED WITHIN THE INSPECTED SCOPE. THESE COUNTS DESCRIBE AVAILABILITY, NOT STUDY QUALITY OR CHECKLIST VALIDITY.

| Question | Eligible settings / sources | Located | Supporting settings and consequence |
|---|---|---|---|
| Protected-condition attack adaptation | 4 / 2 | 2/4 | Communication graphs: transformed captures and live OHTTP. Both come from one preprint. |
| Original-task utility under the same intervention | 4 / 2 | 1/4 | WRA visibility comparison. It does not price WRA's executed-decoy condition. |
| Adaptation and original-task utility in one setting | 4 / 2 | 0/4 | Adaptation and utility are located in different records; transport cost alone does not fill the task-outcome field. |
| Original-task utility for executed cover | 1 / 1 | 0/1 | WRA's five research decoys report recovery and extra research; matched original-task utility is not located. |

4) **Decide using the declared tolerances.** An execution outside its state or resource contract fails that requirement even if its goal passes. An attacker exceeding the allowed inference threshold defeats the corresponding protection claim. Passing a finite attack suite is not a general secrecy proof.

Source-matched examples. WRA-decoy fills the privacy and extra-research fields, while its original-task quality under that intervention is unresolved. WRA-visibility supplies matched quality, but for a different policy. Live OHTTP supplies adapted inference and traffic cost, while original-task utility remains unresolved. ANT supplies observation/label evidence without a protection comparison. These entries preserve each study's scope; they are not paper-quality scores. Supplement S7 gives the complete source-located forms.

## 9. Synthesis and Research Agenda

**The field map turns a privacy score into a protection decision by binding its observations to executable work.** The strongest cross-study consequence is intervention matching: WRA's visibility utility cannot establish executed-decoy utility, and adapted OHTTP inference with traffic cost does not establish original-task preservation. The technical case adds the dispatcher: controller cover can omit model waits and become removable, while planner cover incurs model work. These connections extend the evaluation object; the underlying adaptive-attack and state-correctness principles are established.

### 9.1. Source dependence

The evidence map follows 33 statements in seven comparisons across 20 source-removal scenarios (Supplement S4.4). Strictly retaining inspected published texts leaves WRA's tool-destination recovery and decoy/visibility distinction, plus CHeaT's defined detection/prevention endpoints. The model-view observations, live OHTTP refitting and each named Agent transfer axis depend on preprints. The local cover case adds implementation-specific evidence, not independent confirmation of those source results. This early SoK is useful for fixing interpretation and evaluation conditions; the small direct corpus limits field-wide empirical conclusions.

### 9.2. Tests that would change the decision

Table 10 gives discriminating tests rather than a list of desirable benchmarks. Priority tests are equal-budget cover schedules under production state contracts; independent role/chain attacks on planner-generated cover; discovery and inference on continuous workloads; and length/path-controlled feedback interventions. Business invariants should admit legitimate concurrent updates. Future evidence can support a policy by meeting declared inference and utility thresholds, or reject it by exposing a budget or state violation.

## 10. Validity, Ethics and Availability

**The conclusions apply to the recorded sources and declared observation conditions.** Coverage combines bounded retrieval with citation tracing; failed indices limit

TABLE 10. TESTABLE OPEN QUESTIONS DERIVED FROM THE SYNTHESIS. Q1 SPECIFIES TRANSFER TESTS. Q2 SEPARATES THE FORWARDING MEDIATORS; Q3 EXTENDS THE CONTROLLED COVER STUDY TO APPLICATION CONSTRAINTS; Q4 EXTENDS LIVE DETECTION TO NEW WORKLOADS AND DEPLOYMENTS. EACH GIVES A RESULT THAT COULD SUPPORT OR REVISE THE PROPOSED EXPLANATION.

| Question and unresolved mechanism | Controlled comparison | Discriminating result |
|---|---|---|
| **Q1. Which signals survive execution changes?** Do attacks identify recurring task structure or features tied to its implementation? | Keep task labels and observation rights fixed. Hold out compositions, coordination and client/scaffold versions separately, then jointly. Compare destination, volume/timing, sequence and recipient-visible feature families under declared frozen and adapted training. | Identify a feature family whose inference survives specified changes, or a failure tied to one changed condition. Report per-axis and joint transfer with attempted-run coverage; a pooled average cannot establish invariance. |
| **Q2. How does feedback reach model traffic?** Which part is content, length or later action structure? | Fix initial instructions; project to category-free task fields. Pair length-matched neutral padding with validity-preserving replacement of supplementary content; hold encoding and follow-up policy fixed where possible. Vary record count separately. Compare prefix, first-feedback and complete model views under tool-only and joint protection. | Additional held-out inference after feedback, with controls that exclude the competing path, supports that attribution. Report an effect interval against a prespecified meaningful-effect threshold. A nonsignificant difference alone does not rule out leakage. |
| **Q3. Can cover preserve task semantics?** When should it stop at transport or execute in isolation? | Compare planner-sampled decoy schedules (destination sequence and timing) with marginally matched dispatch under production tool contracts and legitimate concurrent updates. Compare no cover, transport termination and isolated execution under declared resource budgets. Include timing-informed removal and independently captured record times; measure inference, delay, original-task completion, quota, state and planner-visible feedback. | A privacy–utility operating point satisfying prespecified state and task tolerances supports the design. Resource exhaustion, changed business state or impaired completion identifies the constraint it violates. |
| **Q4. Which inference survives task discovery?** Does cross-link association recover information hidden by waits and multiplexing? | Train on one concurrency/load and test new compositions, loads and gateways. Compare tool-only and associated tool/model views with observable flow identifiers; keep ground-truth task and stream mappings outside attacker inputs. Include real unrelated traffic and constant-output detector references. | Measure interval recall, false alarms per unit time and correctly localized inference together. Gains must survive changed workloads and loads at a stated false-alarm budget. |

recall. Most direct sources are preprints. The independent coder's decision-critical subset agrees on the four-setting privacy denominator and its adaptation/utility fields; other extracted fields do not inherit that reliability estimate.

The live case uses synthetic local tools, two fixed 8B models and reused fixtures/phrasings. The original analysis supplies call windows; the new record-only analysis retains episode-associated connections but supplies no call boundaries. Same-service calls serialize. This does not test unknown-task discovery, unrestricted multiplexing or a remote deployment. Minimum historical holds do not ensure achieved timing distributions match. Persistent caches and phase order affect costs. Intervals condition on the fitted attacks and workload; test maxima are descriptive, not all-attacker bounds. Shadow's identical selected predictors do not establish role indistinguishability. The separate forwarding pilot remains inconclusive and changes content, length and order together.

Ethics and assistance. All experimental inputs are synthetic; capture filters restrict collection to the experiment. The adverse-write control runs on isolated fixtures. No customers or production business states are affected. An earlier hosted retest used account-holder authorization; provider research terms remain unverified and its results support no main claim. AI assisted extraction, implementation and writing; measurements are backed by saved observations and executable analysis. Traffic inference can enable profiling. Credentials, transcripts, raw captures, model weights and third-party full texts are excluded from distribution.

Availability. The source release contains the paper, generated aggregate tables and analysis code. The forty-four-setting extraction and source-locator sheets accompany the local review package. Complete individual record metadata and predictions remain in the author's local package; they are not claimed publicly available. Rebuilding the new raw-record inputs needs the retained private records. Aggregate tables, complete configurations and saved-verification scopes provide the alternative checking path.

## 11. Conclusion

Executed cover creates both network observations and application work. This SoK maps the existing evidence to those two outputs and provides a joint protocol for residual inference, task completion, business state and resource cost. The live case shows how a controller's dispatch process exposes cover, how historical and planner-generated schedules change the measured attacks, and why a goal score

alone cannot certify the whole execution. Protection claims become actionable when their privacy and business evidence describe the same policy on the same workload.

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